DEGREE IS ADDITIVE IN RANKS UNDER TENSOR PRODUCT OF LOCALLY FREE MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Both cited lemmas are delivered companions; the final substitution is written out in this spec's third obligation exactly as the author's 'elementary arithmetic' determines it.

Allowed standard prerequisites (editorial).

- Degree as a sum over irreducible components. delivered companion lemma-degree-in-terms-of-components. Reduction to the curve case.
- Chi of a tensor product on a proper curve. delivered companion lemma-degree-on-proper-curve. The formula chi(E tensor F) = r deg(E) + n chi(F).
- Definition of the degree of a locally free module on a curve. chapter definition cited in the proof of lemma-degree-on-proper-curve. deg(M) = chi(M) - rank(M) chi(O_X), used for the final substitution.

Source reference. Original lemma 0AYW: sources/stacks-chow/tags.txt, source lines 13804–13804. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYV: sources/stacks-chow/tags.txt, source lines 13803–13803. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field, let X be a proper scheme of dimension ≤ 1 over k, and let $\mathcal{E}$, $\mathcal{V}$ be locally free $\mathcal{O}_X$-modules of constant finite rank. Then*

$$\deg(\mathcal{E} \otimes \mathcal{V}) = \mathit{rank}(\mathcal{E}) \deg(\mathcal{V}) + \mathit{rank}(\mathcal{V}) \deg(\mathcal{E})$$

Proof. By Lemma 0AYW and elementary arithmetic, we reduce to the case of a proper curve. This case follows from Lemma 0AYV. □

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